

Course Module Topic	Surface Water Engineering	
		Hydraulic Structure Design
	Culvert design	

This handout will provide basic information to design a culvert. A culvert is an artificial water passage under a road, railway or canal, or through an embankment. It is used particularly as a drainage structure and it being used more and more to replace older small bridges.

A. CULVERT DESIGN

I. Design problem

It is necessary to determine the size of a culvert for a given discharge under the constraint of a maximum upstream water level and possibly an imposed downstream water level. We can write for the discharge Q :

$$Q = Q(H, d_t, D, L, S, n, g, G) \quad (1)$$

Where H , d_t , D and L , the geometric variables are defined in Figure 1, S is the culvert slope, n is the Manning's coefficient for the culvert, g is the gravitational acceleration and G represents the inlet geometry

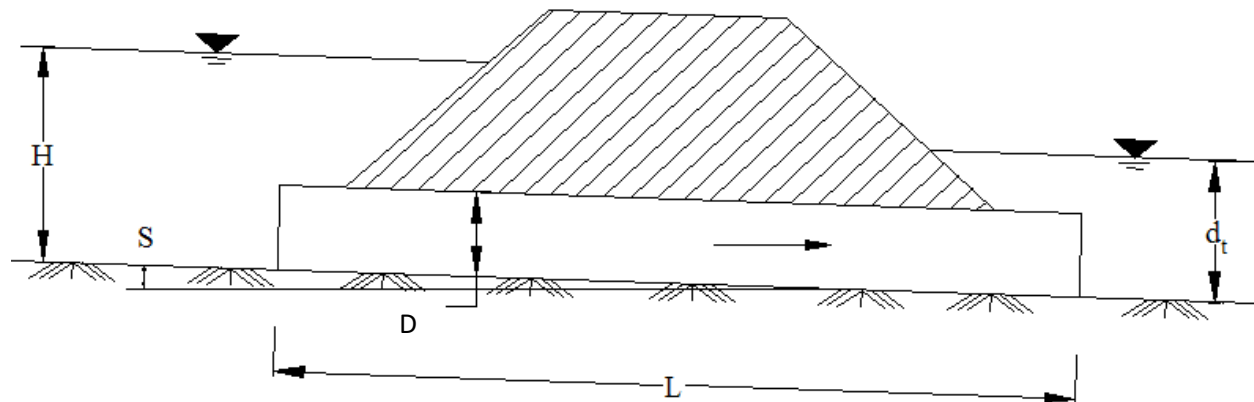


Figure 1. Culvert – Definition of Symbols

Notes:

1. Maximum upstream depth H is set by the embankment height to avoid overtopping or by upstream banks to avoid flooding.
2. The tailwater depth d_t is set by the stream channel characteristics or by whatever energy dissipation device is built adjacent to the culvert.
3. For a circular culvert D is the diameter but for other shapes (square, rectangular, arch) a representative length is used for D .
4. Manning's n ranges from 0.011 for smooth concrete culvert to 0.035 for a culvert with internal ribs.
5. Inlet geometry varies considerably. Some common types are shown in Fig.2.

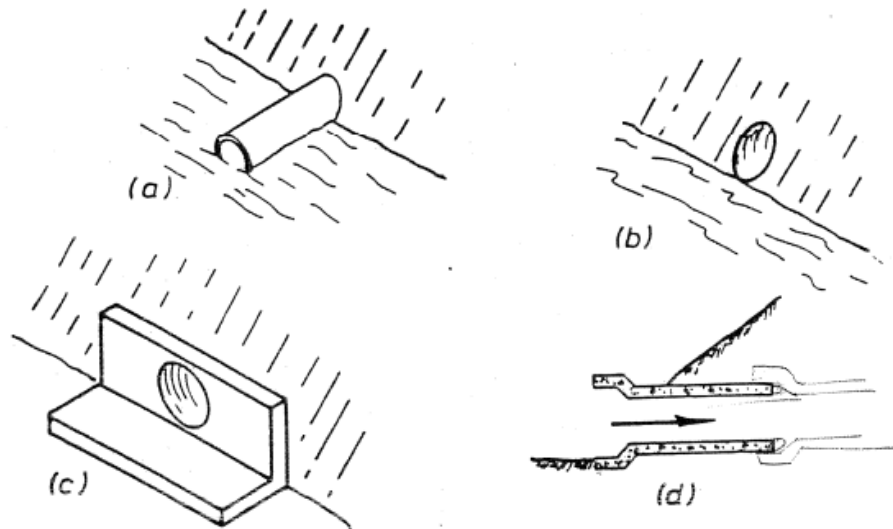


Figure 2. Inlet Geometries: (a) Projecting (b) Mitred Inlet (c) Headwall Inlet (d) Socket Inlet
(Source: Culvert Manual)

6. The parameters of Equation (1) combine to determine flow conditions within the culvert. Two basic types occur – Inlet control and Outlet control.

II. Inlet control

Flow is essentially orifice flow with the culvert flowing partly full throughout its whole length, see Figure 3.

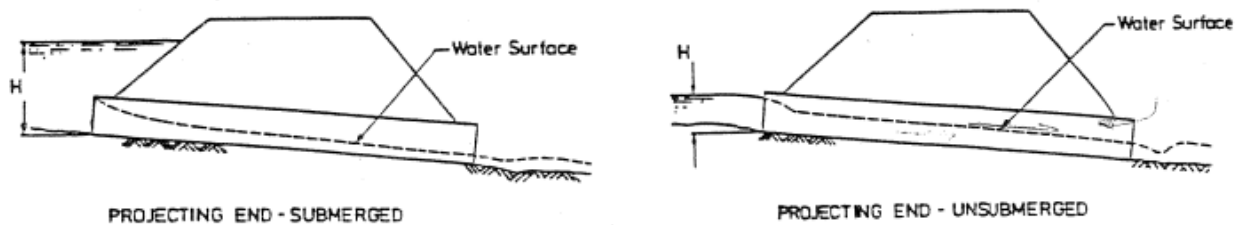


Figure 3. Inlet controlled culvert flow (Source: Culvert Manual)

Requirements for inlet control are that the culvert is

- (i) Short enough and/or
- (ii) Steep enough and/or
- (iii) Smooth enough and
- (iv) $d_t < D$ to allow the air demand to be met.

For inlet control Equation (1) becomes

$$Q = Q(H, D, g, G) \quad (2)$$

Three cases are clearly distinguished:

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1. $H > 1.5D$ this is true orifice flow for which

$$Q = CA\sqrt{2g(H - \frac{D}{2})} \quad (3)$$

Or
$$\frac{Q}{D^2\sqrt{gD}} = C \frac{A}{D^2} \sqrt{2\left(\frac{H}{D} - \frac{1}{2}\right)} = f\left(\frac{H}{D}, G\right) \quad (4)$$

Where the discharge coefficient C represents the inlet geometry and A is the cross sectional area of the culvert.

Equation (4) is plotted for $H/D > 1.5$ and circular culverts ($A/D^2=0.785$) in Figure 4. Values of the coefficient C used are listed in Table 1 and may be used to extend the range of Equation 4 to large values of H/D .

Table 1. Coefficients used for Inlet Control

Curve, Figure 4	A	B	C	D	E
Value of C , Eq.(4)	0.5	0.6	0.7	0.75	0.97

2. $H < D$ this is a suppressed weir flow with the culvert size being less than that of the approaching stream. The appropriate equation has the form

$$Q = kH^r \quad (5)$$

Where k, r reflect the culvert size and entry geometry. Model tests have been used to derive equations having the form of Equation (4).

$$\frac{Q}{D^2\sqrt{gD}} = f\left(\frac{H}{D}, G\right) \quad (6)$$

These are plotted in Figure 4.

3. $D < H < 1.5D$ this is a transition between weir and orifice flow. Model studies have provided the transition curves shown Figure 4.

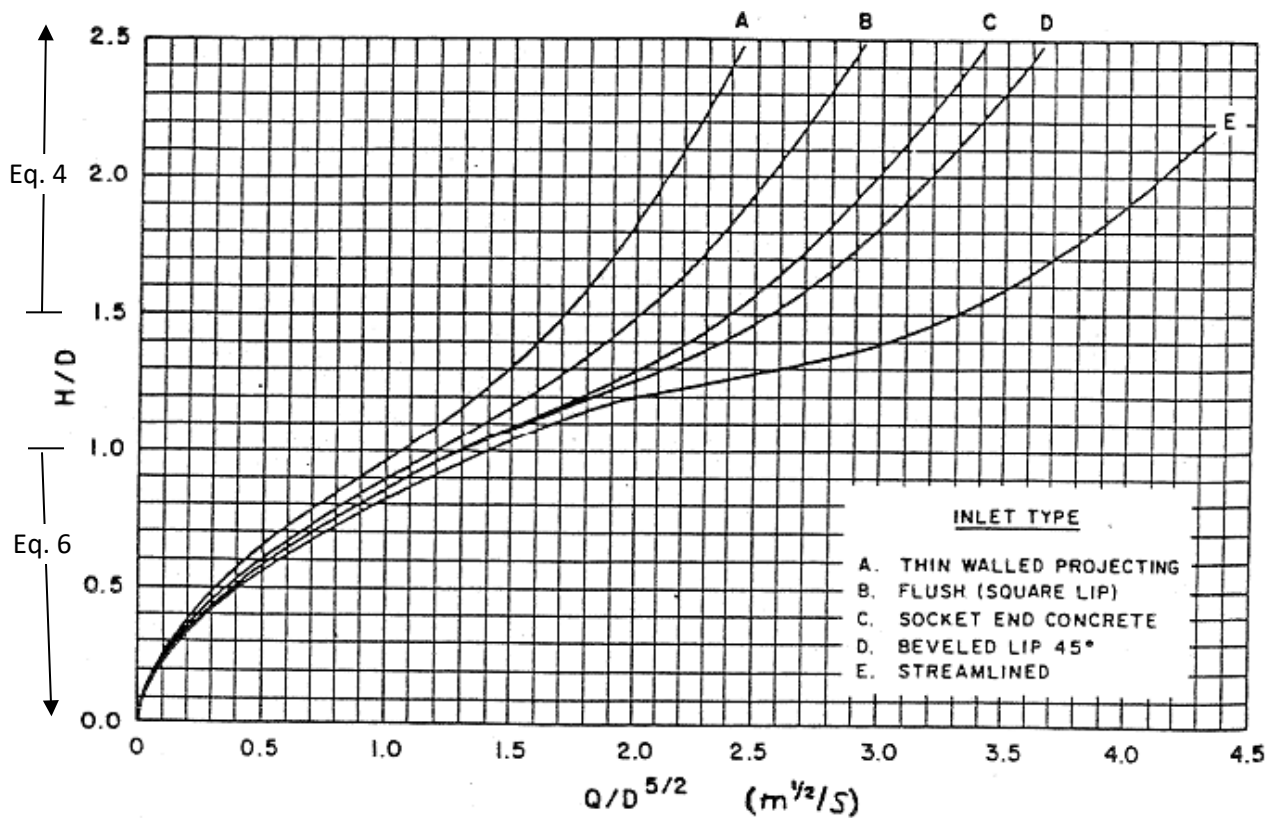


Figure 4. Rating curves for Inlet Controlled Flow through Circular Culverts of Diameter D

Note: In using Figure 4 to size a culvert a trial and error solution is necessary. As a useful starting point assume $A = Q/3$ in SI units. This converts to $D = 0.65\sqrt{Q}$ for use in SI units.

III. Outlet control

Culverts flowing full over at least part of their length or those with flow depths controlled by friction within the culvert are no longer inlet controlled. This happens as S decreases or as L , n or d_t increase. The discharge problem then reverts to Equation (1).

$$Q = (H, d_t, D, L, S, n, g, G) \quad (7)$$

With a number of different possibilities for flow configuration (Figure 5).

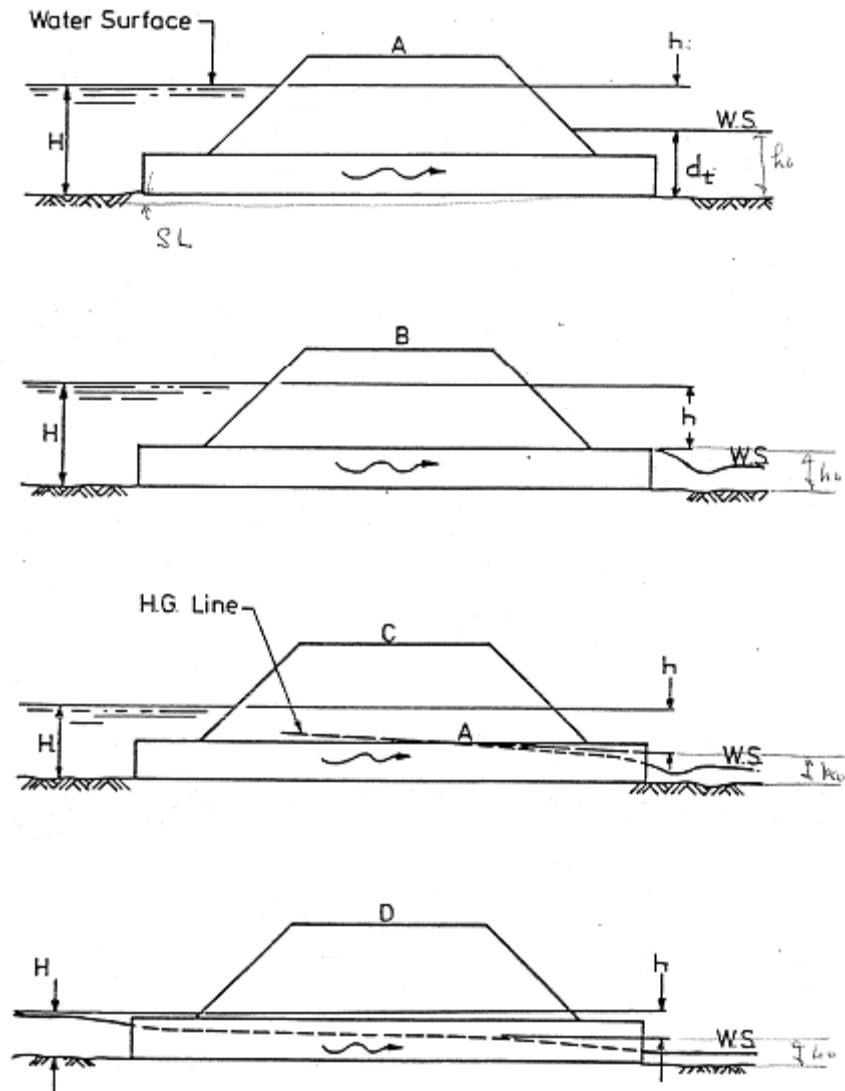


Figure 5. Possible outlet controlled flows (Source: Culvert Manual)

1. Full flow throughout – Case A, B Figure 5.
This is pipe flow and one writes for the head loss h across the culvert

$$h = \left[K_c + \frac{2gn^2L}{R^{4/3}} + 1 \right] \frac{V^2}{2g}$$

$$= H + SL - d_t \text{ for case A} \quad (6)$$

$$= H + SL - D \text{ for case B} \quad (7)$$

Where R is the hydraulic radius of the culvert and K_c is an entrance loss coefficient (Table 2).

Table 2. Entrance loss coefficient K_c

<u>Concrete Pipe: projecting and headwall</u>	
Socket End	0.2
Square Cut End	0.5
<u>Concrete Box</u>	
Headwall, square edge	0.5
Headwall, rounded edge	0.2
Wingwalls, parallel	0.7
$10^\circ - 30^\circ$	0.5
$30^\circ - 75^\circ$	0.4

- Notes: a. A solution for D requires trial and error
b. Case A will occur whenever $d_t > D$
c. Case B will occur whenever $\frac{Q}{D^2\sqrt{gD}} > 1.1$, provided d_t is small enough not to effect the culvert flow. This condition corresponds to critical depth equaling 0.96 of the culvert diameter.
d. For both case A and B
- $$h = H + SL - h_o \quad (8)$$
- where h_o is the elevation of the hydraulic grade line at the culvert exit.

2. Full flow – Part of length – Case C, Figure 5

When critical depth falls below the culvert soffit the water surface will be as shown in C of Figure 5. The hydraulic grade line will pass through the point (A in Figure 5, Case C) at which the water surface leaves the soffit. Calculations show that the hydraulic grade line will reach the end of the culvert at an elevation $h_o = (d_c + D)/2$ where d_c is the critical depth. Thus, using the form of Equation (8)

$$h = H + SL - (d_c + D)/2 \quad (9)$$

The critical depth d_c can be obtained from Figure 6.

Should the tail water depth d_t exceed $(d_c + D)/2$ then the culvert flow will be influenced by the tailwater. In this case use $h_o = d_t$ and thus

$$h = H + SL - d_t \quad (10)$$

For case C to occur there must be sufficient upstream energy to supply the velocity head in the culvert and to overcome the entry losses. Hence

$$H > D + (1 + K_e) \frac{V^2}{2g} \quad (11)$$

If this is not so then case D occurs with part full flow throughout (see 3 below).

3. Part full flow throughout – Case D, Figure 5.

Accurate head water depths H can only be determined by a full backwater computation. Approximate values may be found by using $h_o = (d_c + D)/2$ or d_t as described above for case C. Values will be satisfactory if $H > 0.75D$.

If $H \leq 0.75 D$ then capacity design charts must be used.

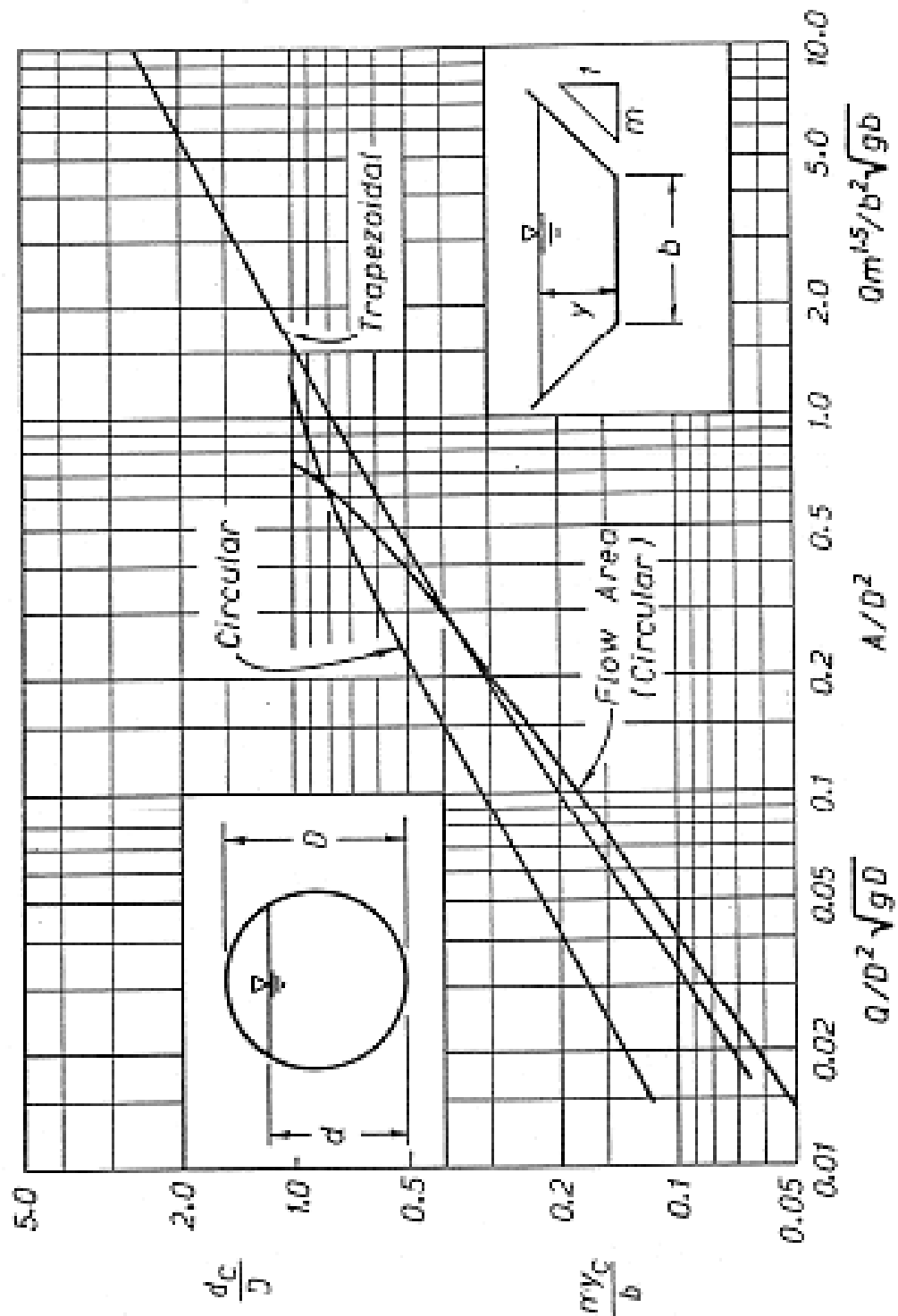


Figure 6. Critical depth curves for circular and trapezoidal channels

4. Determination of Head Loss h

Equations (6), (7), (9) and (10) require the head loss h to be known before the upstream water level can be calculated. Nomographs have been prepared for particular cases, see examples in Figure 8 and 9. To use the nomographs the following steps, as illustrated in Figure 7 must be followed.

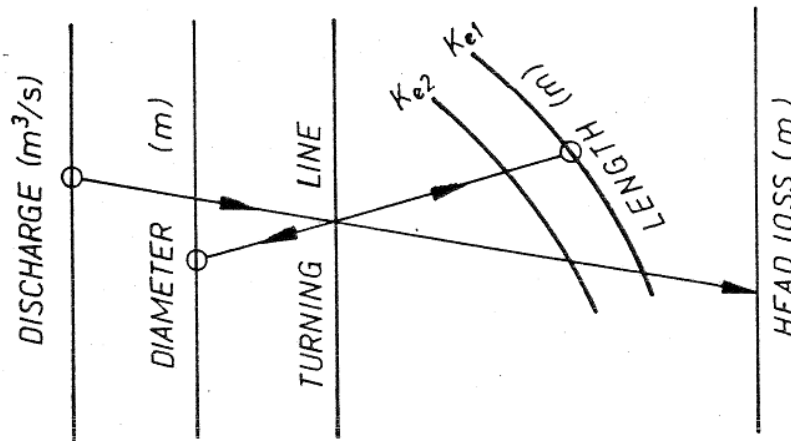


Figure 7. Use of outlet control nomograph

1. Select an appropriate entry coefficient K_e from Table 2.
2. Select nomograph corresponding to culvert type, shape and roughness. If Manning's n for the culvert differs from that of the chart calculate an adjusted length L' using

$$L' = \left(\frac{n}{n'}\right)^2 L \quad (12)$$

Where L is the culvert length, n is the culvert roughness and n' is the value of Manning's n on the chart. Note that this implies n^2L is constant.

3. Join L' on the appropriate K_e curve to the culvert diameter D and mark turning line. Interpolation between K_e curves may be necessary.
4. Join discharge Q to mark on turning line and extend to h .

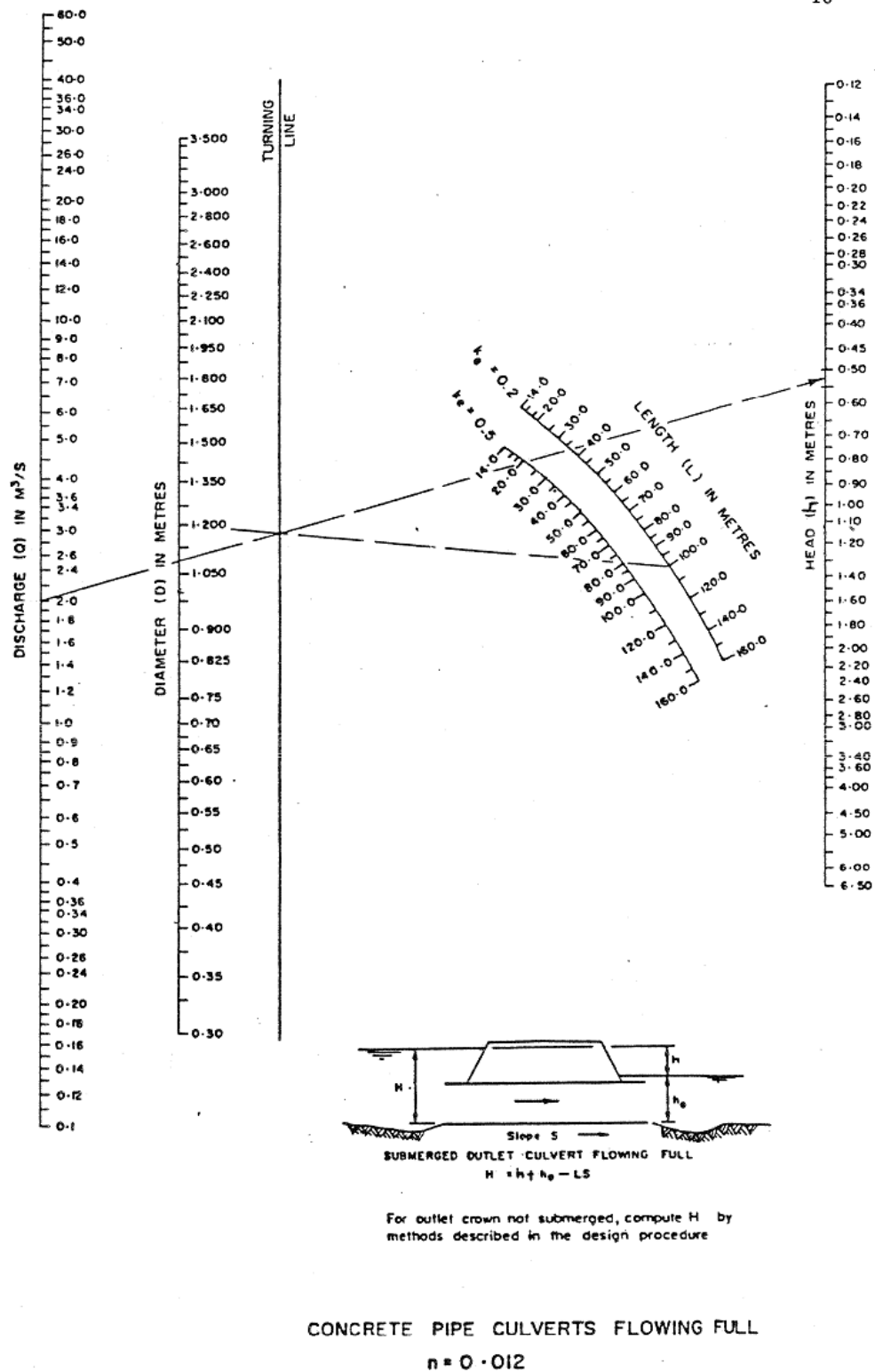
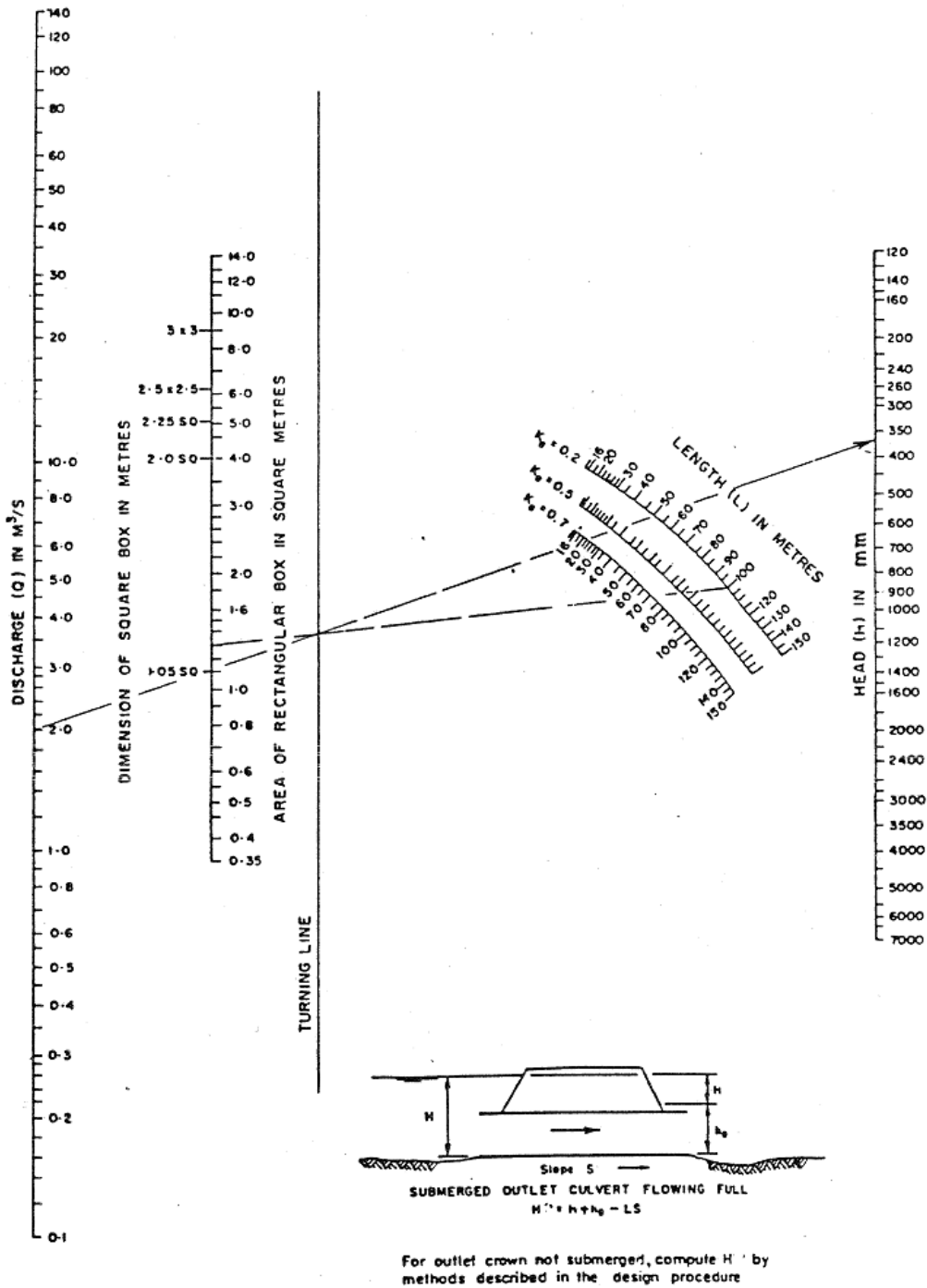


Figure 8. Nomograph for head loss h , under Outlet Control – Circular Culverts



CONCRETE BOX CULVERTS FLOWING FULL
 $n = 0.012$

Figure 9. Nomograph for head loss h , under Outlet Control – Box Culverts

Example: Find a suitable culvert diameter for a flow of $2.3 \text{ m}^3/\text{s}$ with a tail water depth of 1.5m, culvert length of 40m on slope 0.002 and allowable head water of 1.8m. Assume $K_c = 0.2$ and $n = 0.013$.

Calculate new length $L' = \left(\frac{0.013}{0.012}\right)^2 40 = 46.9 \text{ m}$

- (i) Assume $D = 1.0\text{m}$ and from Figure 8, $h = 1.3\text{m}$
 With $d_t = 1.5 > D$ equation (6) is appropriate, thus
 $h = H + SL - d_t$
 So $H = 1.3 - 0.002 \times 40 + 1.5 = 2.72 \text{ m}$
- (ii) Assume $D = 1.35\text{m}$ thus $h = 0.28 \text{ m}$ (Figure 8)
 So $H = 0.28 - 0.002 \times 40 + 1.5 = 1.70 \text{ m}$.
 Hence use a 1.35m diameter culvert.

IV. Inlet control nomographs

Nomographs have been prepared for inlet control conditions. Two examples are given in Figure 10 and 11. To size a culvert one assumes an initial size, joins this size to the known flow, extends to the headwater depth ratio line 1 and then moves horizontally to the scale corresponding to the entry condition. In Figure 10, $D = 1.2 \text{ m}$, $Q = 5.0 \text{ m}^3/\text{s}$ and $H/D = 2.35$ for a groove end projecting culvert.

Note: groove and projecting, Figure 10, corresponding to curve C, socket end of Figure 4, and square edge with headwall, Figure 10, corresponds to curve B, flush square lip of Figure 4.

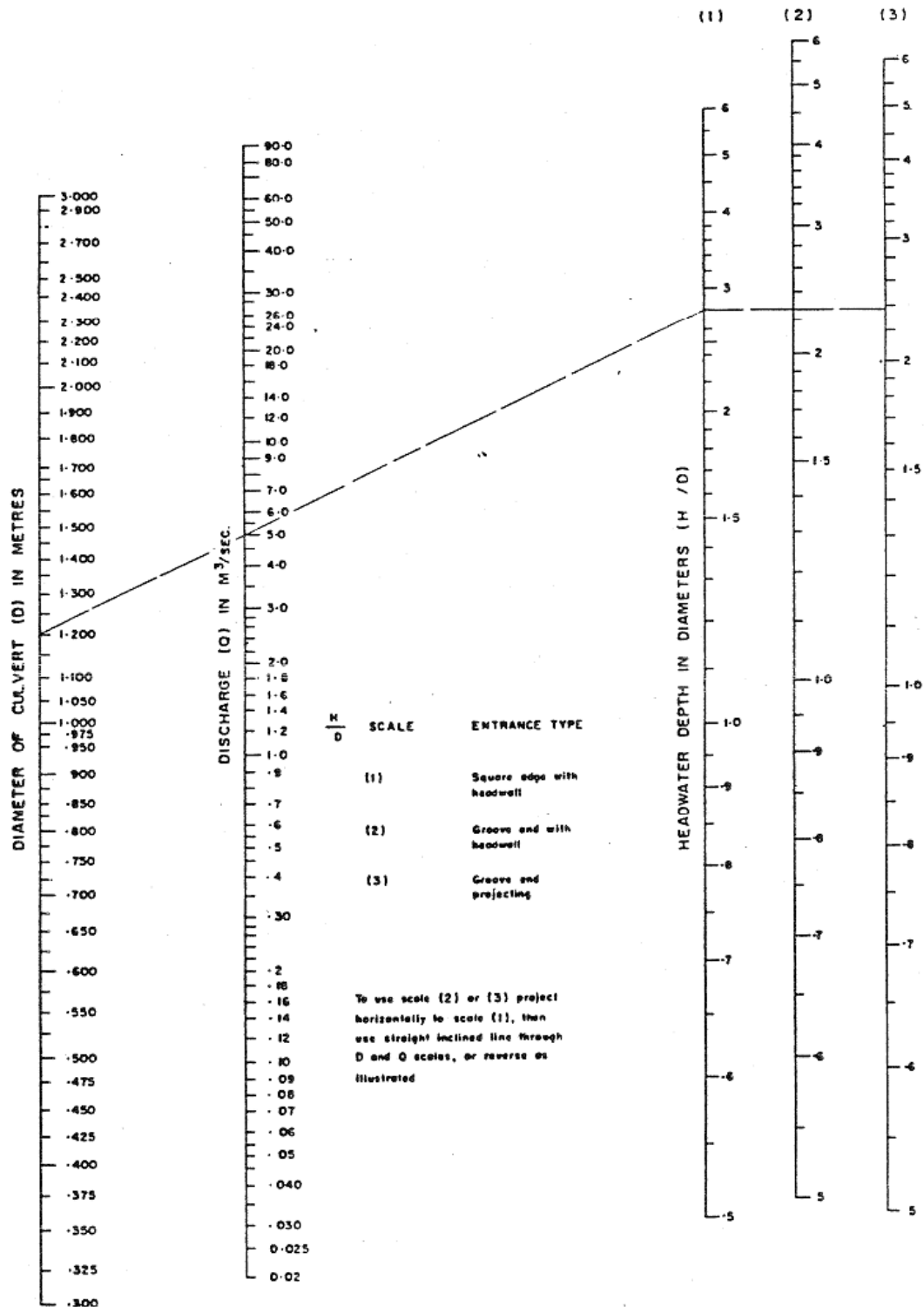


Figure 10. Headwater depth for circular pipe culverts with inlet control

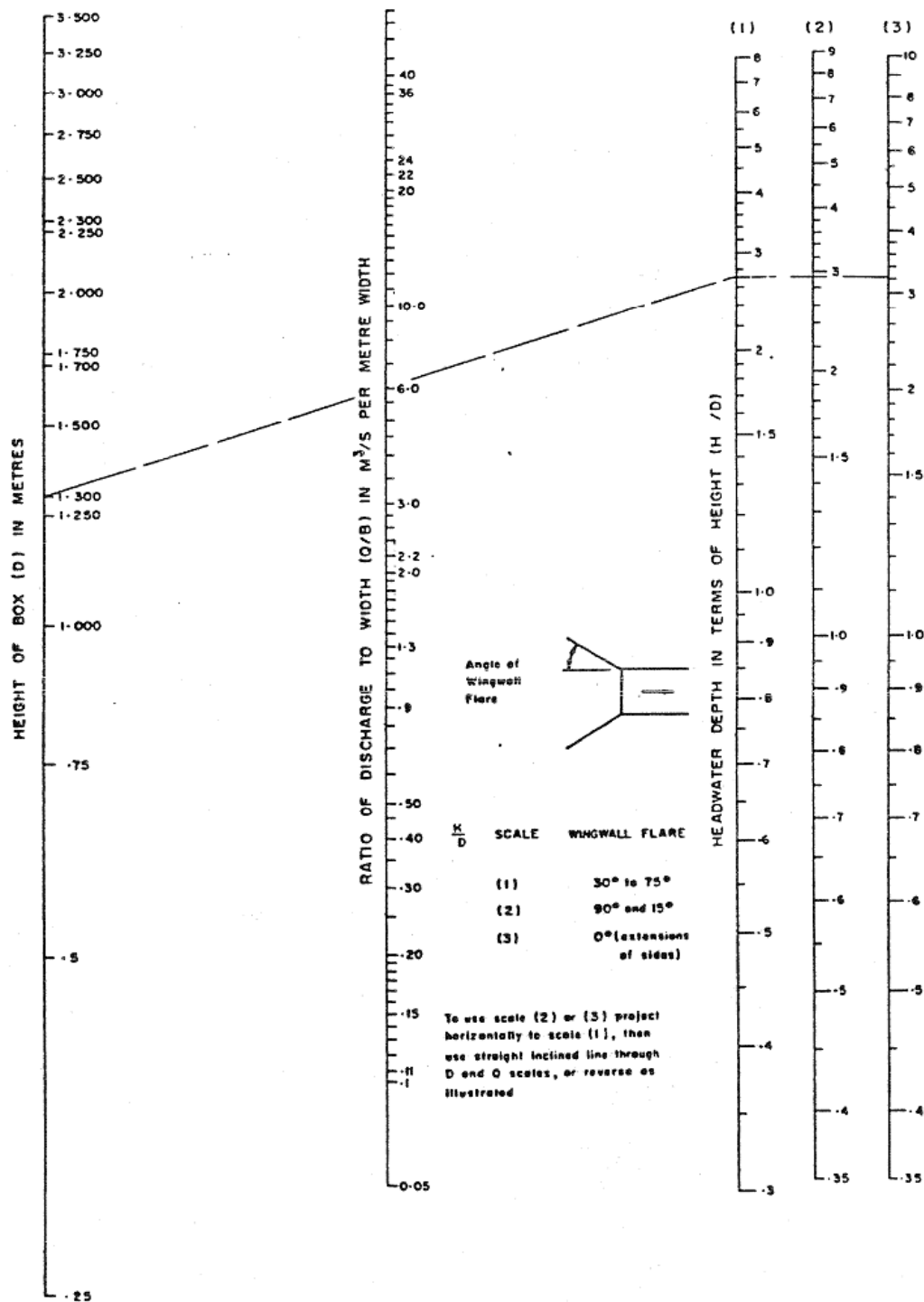


Figure 11. Headwater depth for box culverts with inlet control

To select culvert size using nomographs, apply the following steps:

- (i) Assume inlet control. For the type of culvert selected use a trial size to determine headwater depth from inlet control nomograph. Compared headwater depth with allowable depth and adjust trial culvert size if necessary until the headwater depth is acceptable. Let this depth be H_i .
- (ii) Assume outlet control and culvert size found in 1 above.
- (iii) Derive tail water depth for the design flood.
- (iv) Determine head loss h from appropriate nomograph and find headwater for outlet control H_o from:

$$H_o = h - SL + h_o \quad (13)$$

$$\text{Where if } d_t \geq D \quad h_o = d_t$$

$$\text{If } d_t < D \quad h_o = \text{large of } \left(\frac{d_c + D}{2}, d_t \right)$$

- (v) If $H_i > H_o$ use trial size and inlet governs

$H_i < H_o$ outlet governs and a large size may be needed. Repeat 4 above with a large size until H_o is less than allowable head water depth. This will be the required size.

V. Design capacity charts

A series of design chart are presented in Figures 13 to 19. Each chart applies to a particular culvert type with charts available for:

- (a) Concrete box culverts with various wing wall angles,
- (b) Concrete pipe culverts, square edged and pipe socket inlet,
- (c) Corrugated metal circular pipe, projecting inlet and head wall inlet,
- (d) Structural plate circular corrugated metal pipe, projecting inlet and head wall inlet,
- (e) Structural plate pipe-arch corrugated culverts, projecting inlet and headwall inlet.

Each chart shows headwater depth as a function of discharge for a given culvert type and size D , and for two values of $(L/100S)$ see Figure 12.

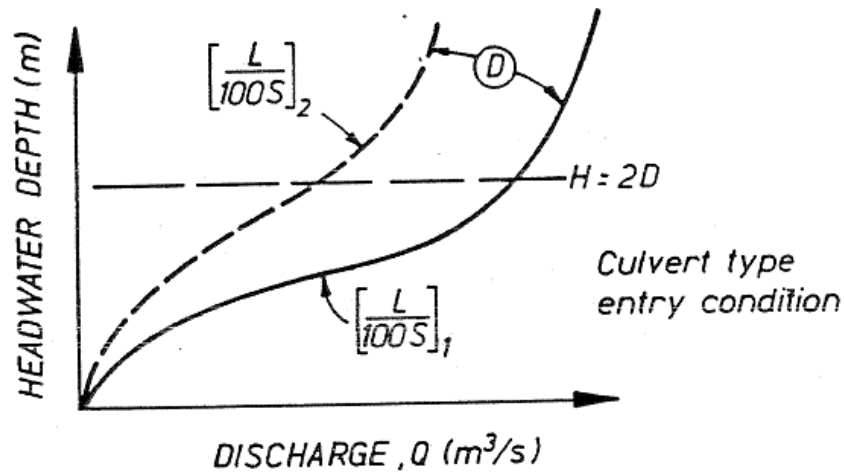


Figure 12. Typical design capacity chart

In Figure 12 $(L/100S)_1$, the solid line, is the upper limit for inlet control. This gives the minimum value of H for a given (Q, D) . Smaller values of $(L/100S)_1$ will mean inlet control and the same value of H .

The dashed line $(L/100S)_2$ is within the outlet control regime and is used to interpolate values of $(L/100S)$. It has been assumed that $d_t < d_c$ and the H values include both entry loss and loss within the culvert. If only a solid line is shown for a given D then the dashed line may be assumed coincidental with it.

Selected design capacity charts are given as Figures 13 to 19.

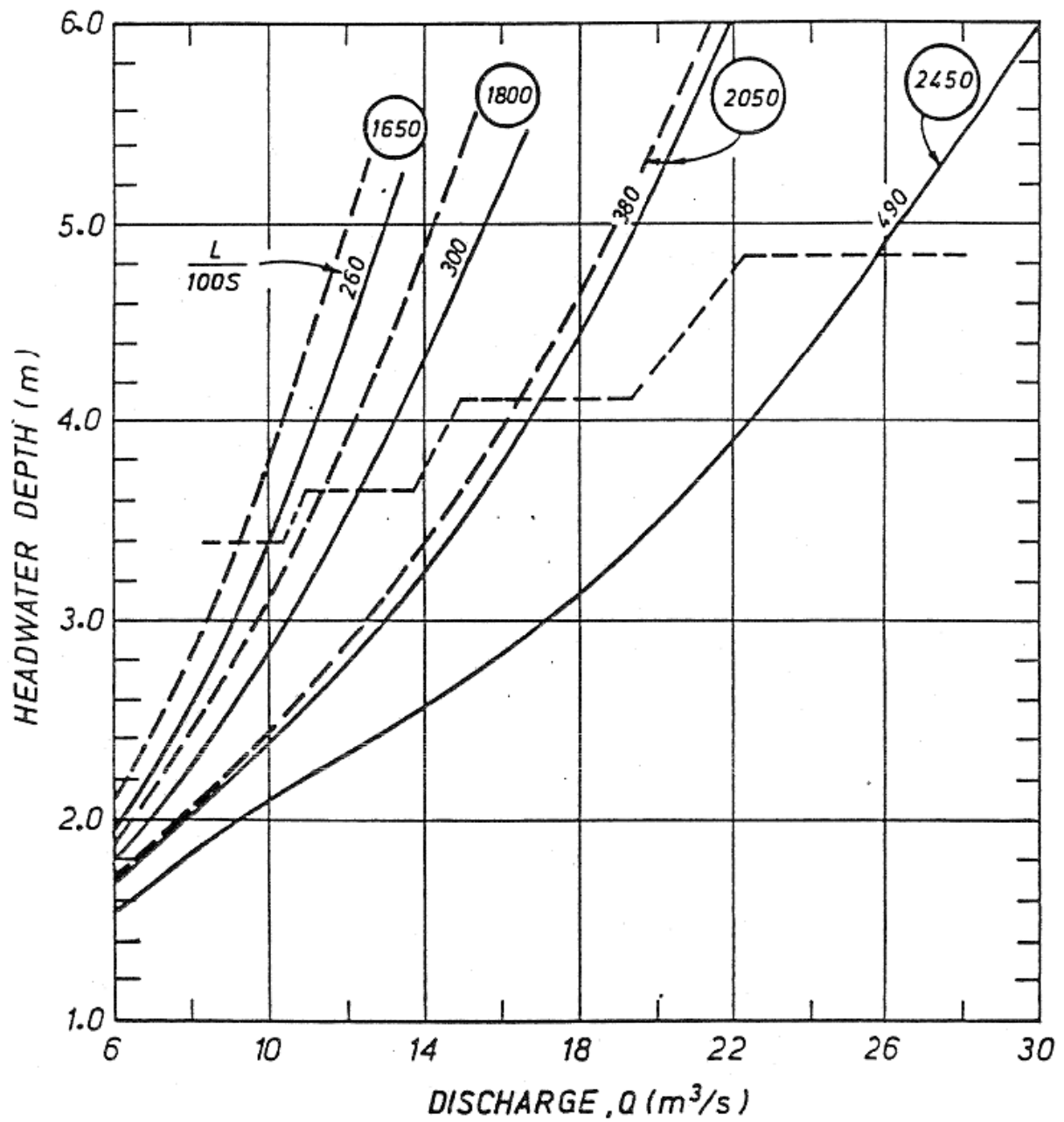


Figure 13. Culvert capacity chart, circular concrete pipes, socket entrance

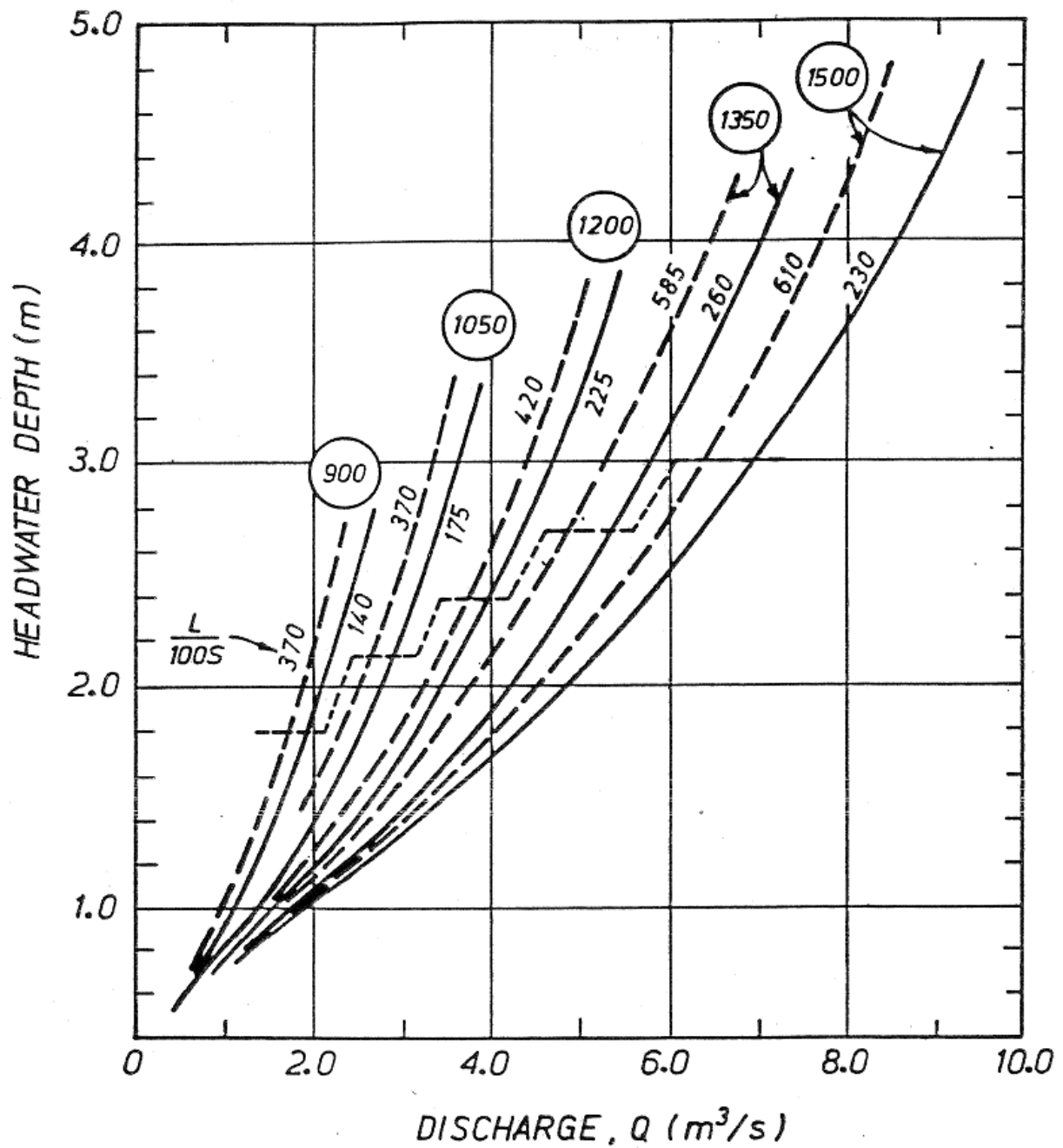


Figure 14. Culvert capacity chart circular concrete pipe, square edged entrance

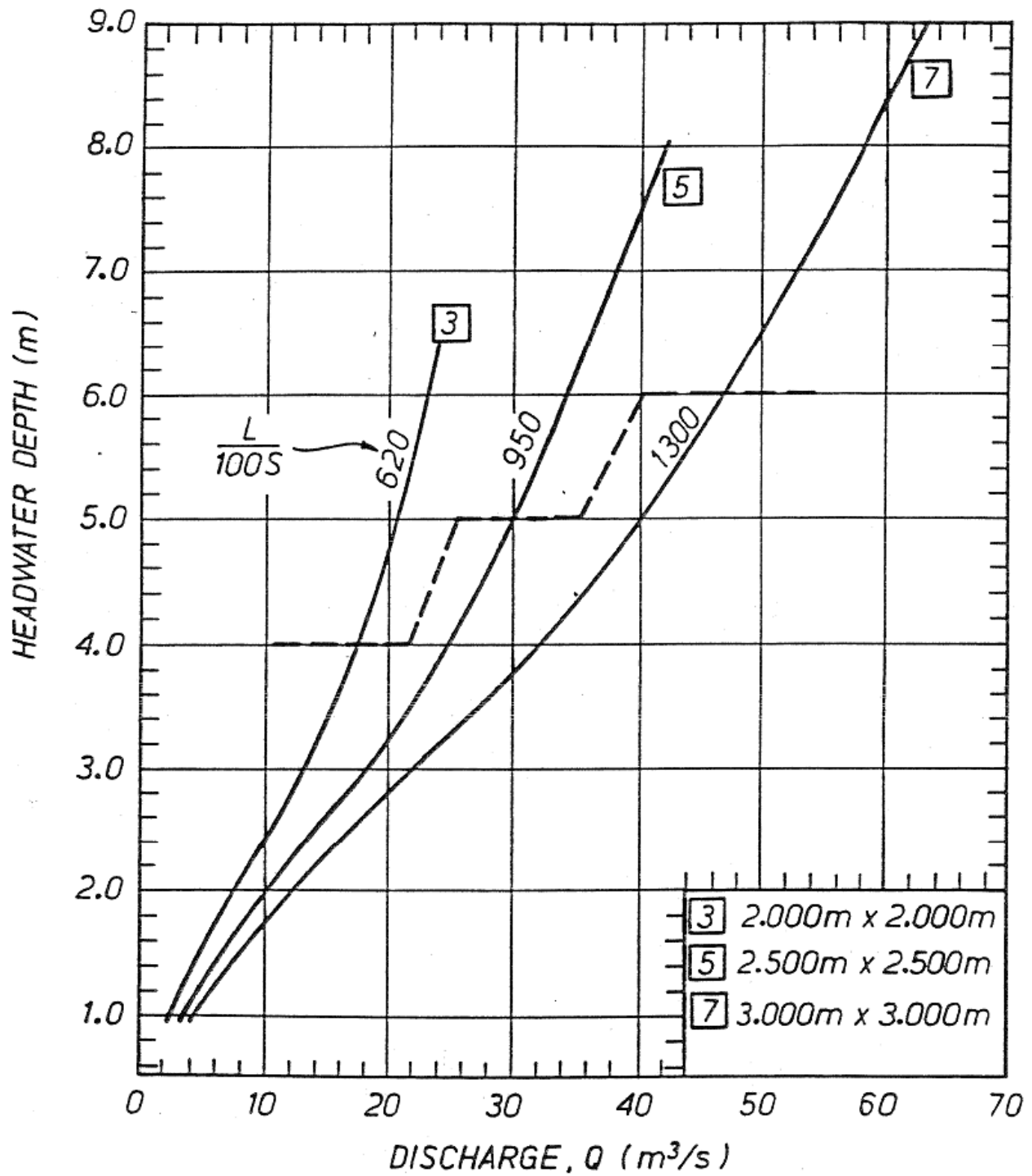


Figure 15. Culvert capacity chart, square concrete box, 90° and 15° wingwalls

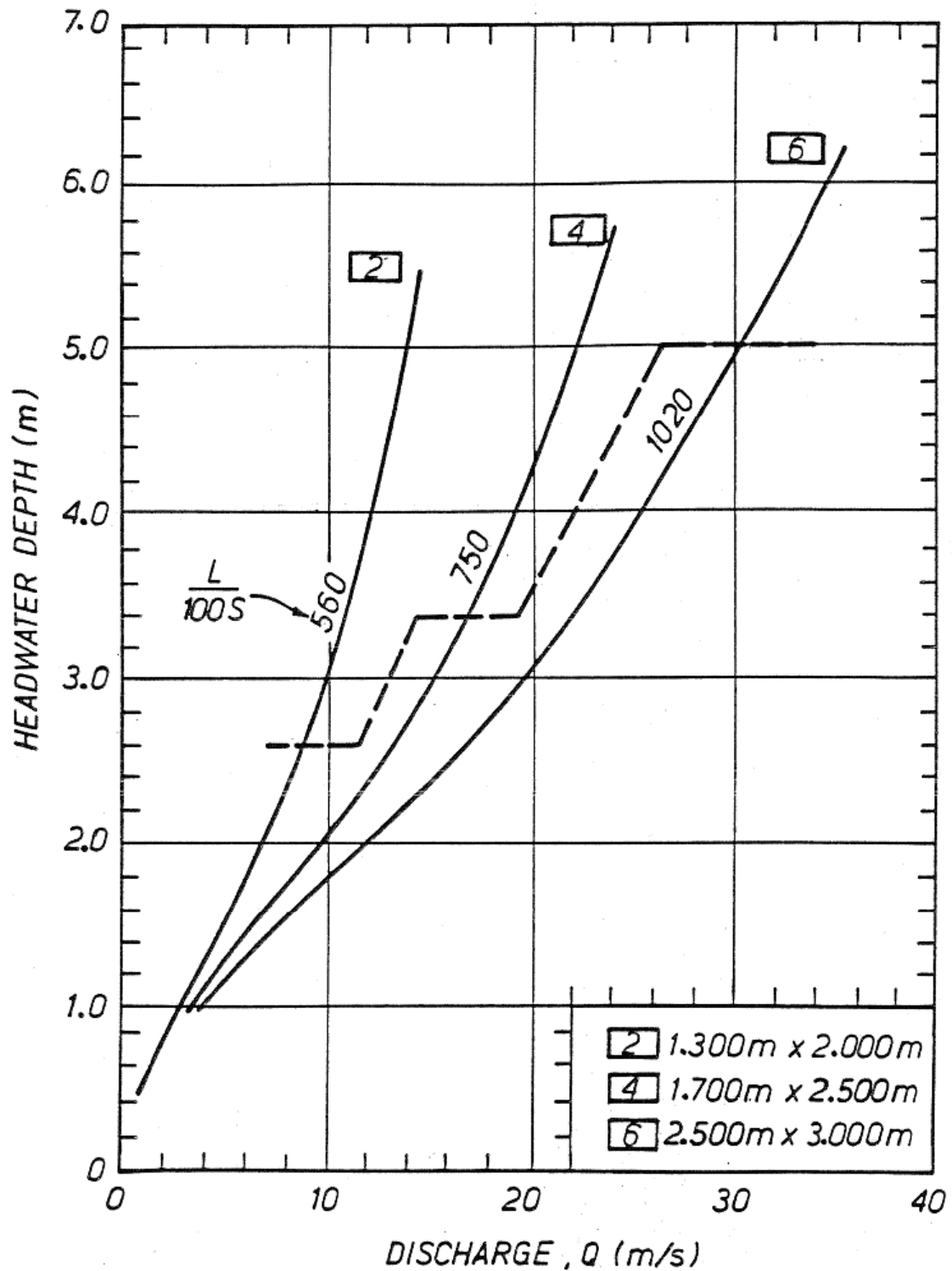


Figure 16. Culvert capacity chart, rectangular concrete box, 90° and 15° wingwalls

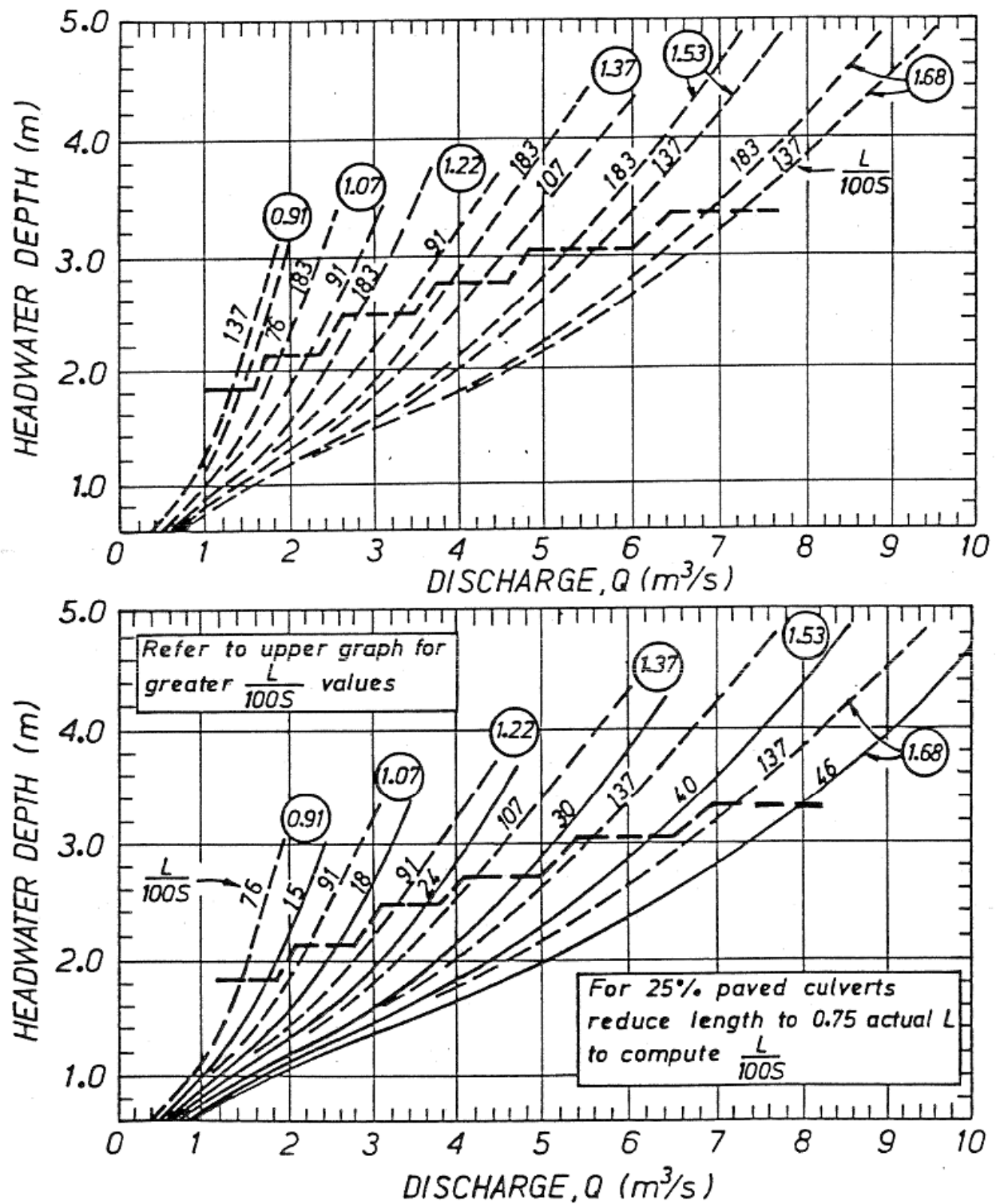


Figure 17. Culvert capacity chart, standard circular corrugated metal pipe, projecting entrance

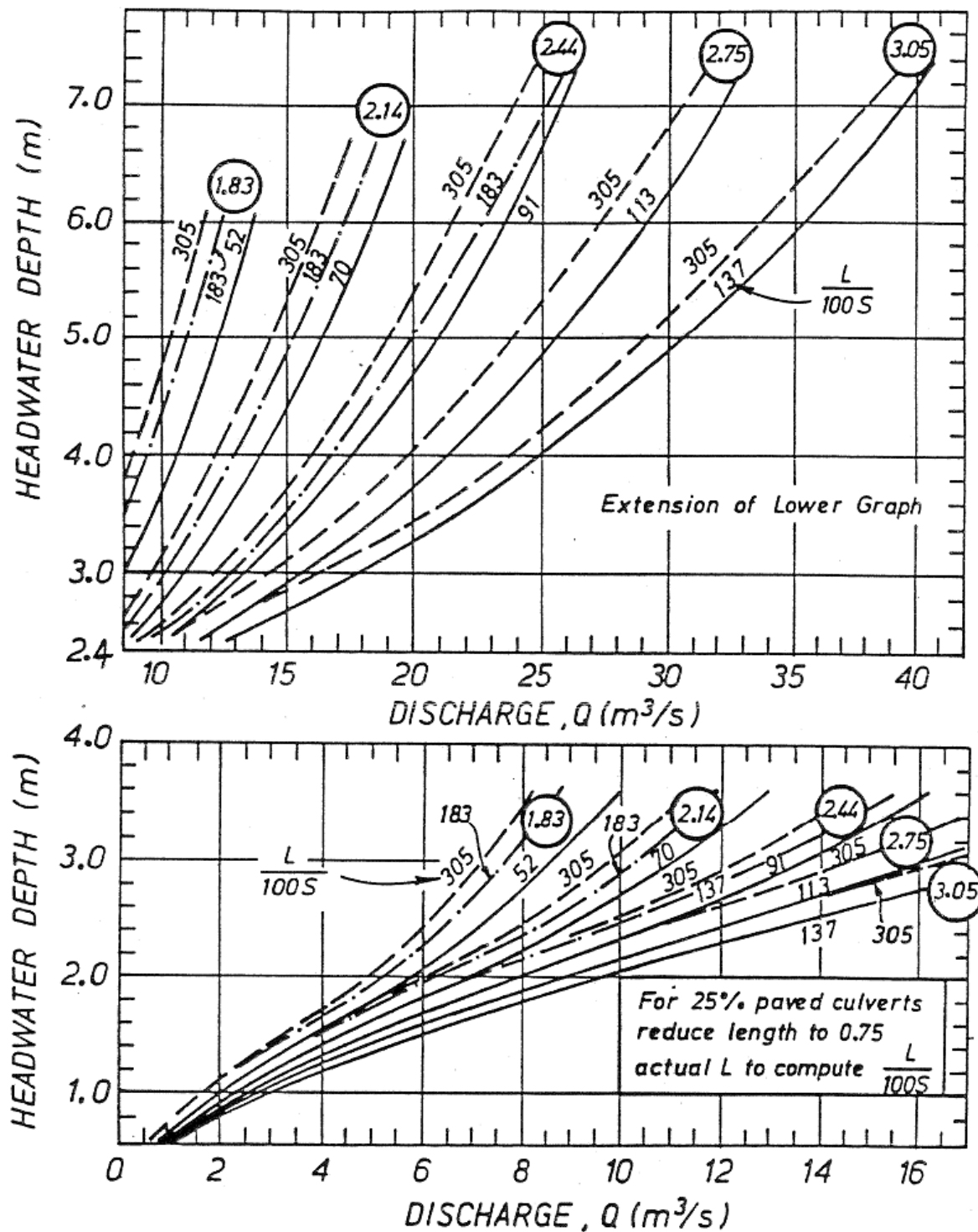


Figure 18. Culvert capacity chart, standard circular corrugated metal pipe, headwall entrance

- 1.1. Culvert type: The culvert type must be matched together with its entry conditions as closely as possible to one of the charts.
- 1.2. Culvert size: interpolation between results obtained using the next larger and next smaller size is permissible.
- 1.3. Culvert slope: the invert of the channel and culvert is assumed to be a straight line. New culverts should have minimum slopes as follows

$$S_{\min} = 0.002 \quad \text{all concrete culverts}$$

$$= 0.004 \quad \text{standard corrugated metal and pipe arch culverts}$$

$$= 0.01 \quad \text{structural corrugated metal and pipe-arch culverts.}$$

For existing culverts with slopes S , less than $S_{\min}$ use $S_{\min}$ and reduce allowable headwater by $L(S_{\min} - S)$.
- 1.4. Large values of $(L/100S)$:
 - (a) For $(L/100S) < 1.5(L/100S)_2$ extrapolate using the solid and dashed lines of constant $(L/100S)$ to find H . If $H < 1.3D$ then the solution can be accepted.
 - (b) $(L/100S) > 1.5(L/100S)_2$ then one must use the nomograph method.
- 1.5. Large values of H :

When $H > 2D$ the values become unreliable and the nomograph method should be used. Lines of $H = 2D$ appear on each chart.
- 1.6. Effects of tailwater levels:

The charts assume the tailwater level is less than the critical depth. If this is not the case then

 - (a) $d_c < d_t < D$. The headwater level H_{chart} should be determined using the appropriate chart and then modified to give headwater level H using

$$H = H_{\text{chart}} + 0.5(d_t - d_c)$$

Provided $(L/100S) \geq 0.5(L/100S)_1$, otherwise $H = H_{\text{chart}}$.
 - (b) $d_t > D$. The nomograph method must be used.

B. DOWNSTREAM PROTECTION

1. The erosion potential created by a culvert must always be checked. Culverts acting with inlet control frequently have exit velocities much in excess of that in the channel into which they discharge. Transition structures and/or erosion control devices may be necessary to ensure the safety of the culvert. Typical measures include a rip rap basin, an apron with artificial roughness elements and basins designed to induce a hydraulic jump. The last mentioned are often referred to as energy dissipaters.

The design procedure for downstream protection is shown in Figure 20 which is based on a similar figure in the culvert manual. The boxes in heavy outline are now discussed.

2. Downstream conditions

Two quantities are of importance

- (i) The permissible velocity in the channel, and
- (ii) The tailwater depth.

Permissible velocity

The materials forming the bed and banks of the downstream channel will determine the non-erosive velocity required in the channel. Tables and charts giving permissible velocities as functions of bed material type, size and classification, type and amount of sediment transport, flood frequency, flow depth, channel alignment and bank slope angle can be found in the culvert manual and other hydraulic design manuals. Typical values are 1m/s for a flow with suspended load over a bed of 1mm particles and 2m/s with bed load and 20mm particles.

Tailwater Depth

Use of Manning's equation is recommended to determine flow depths for the design flood. The resultant velocity must be less than the permissible velocity for the channel to be stable. Typical Manning's n values are: 0.02 for clean unlined channels in sand and gravel, 0.035 for winding natural streams and canals with rough stony beds, 0.06 for sluggish rivers with weedy or vegetated reaches.

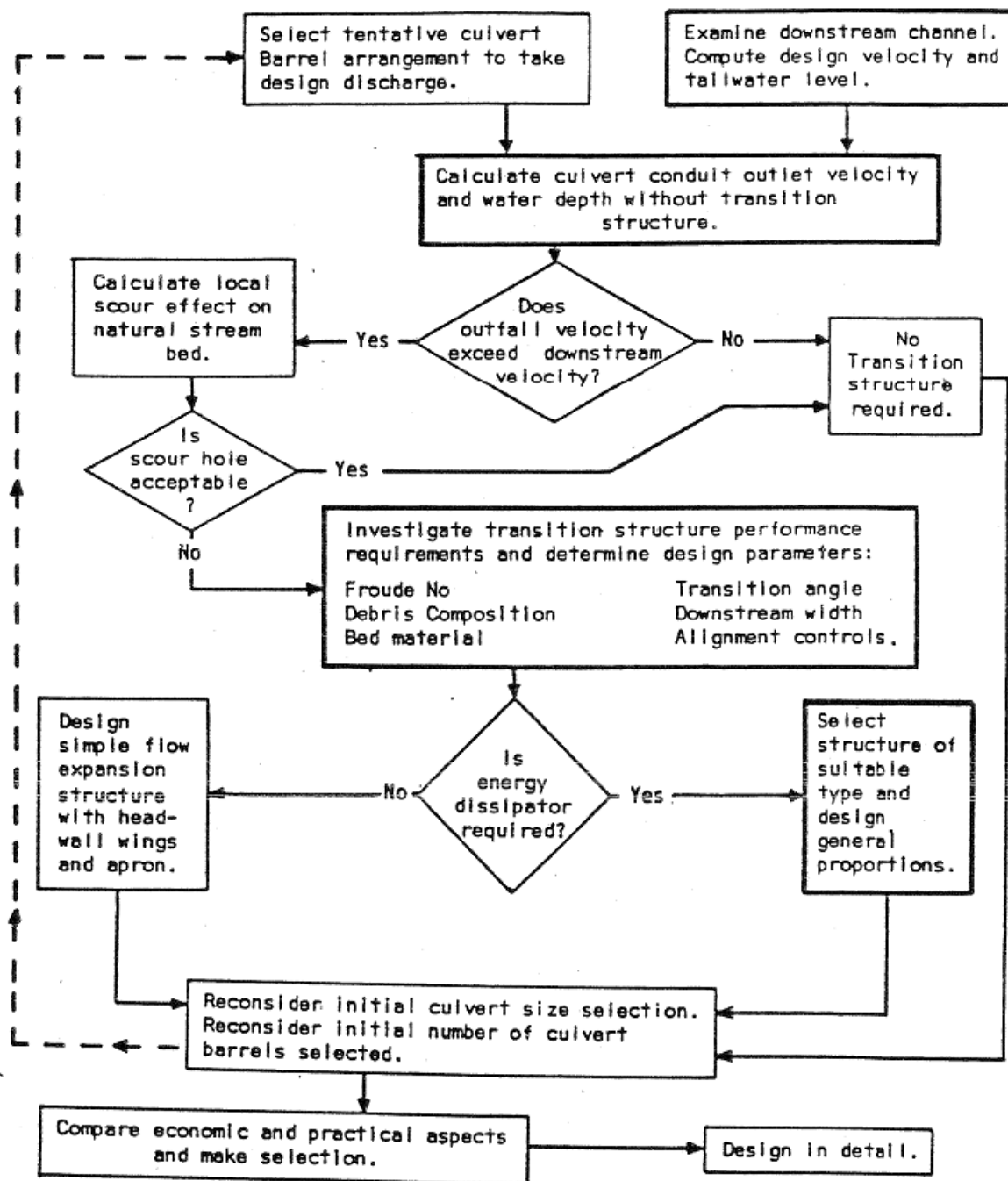


Figure 20. Design procedure for downstream protection

3. Outlet velocity

Culverts with outlet control – If the outlet is submerged (Case A, Figure 5) the outlet velocity is the design discharge divided by the culvert area. The submerged jet from the culvert will rapidly dissipate and erosion will not normally be a problem. At lower flows when the tail water level drops below the culvert soffit there may be erosion problems and such cases should be checked.

- Case B, Figure 5 will have an outlet velocity based on the full culvert area.
- Case C, D of Figure 5 will have the outlet velocity based on the wetted area as determined by the end depth. This depth is affected by the tailwater depth and may be determined using Figure 21 for rectangular and circular culverts.

Note that for zero tailwater depth, the end depth y_e is less than y_c . This is a result of the curved streamlines near the exit forcing critical depth to occur some distance upstream (5 to 7 flow depths in a rectangular channel).

Example: Let $Q = 1.7 \text{ m}^3/\text{s}$, $D = 1500 \text{ mm}$ and tailwater depth = 0.6 m. From Figure 6 $y_c = 0.675 \text{ m}$. From Figure 21 $y_e = 0.615 \text{ m}$. Using Figure 6 $A/D^2 = 0.31$ and thus $A = 0.698 \text{ m}^2$ and $V = Q/A = 2.44 \text{ m/s}$.

Culverts with inlet control – Flow in the culvert will be at normal depth and thus Mannings equation is appropriate. Use Table 3 to determine the normal depth by first calculating $Q_n/(D^{8/3}S^{1/2})$. The exit velocity is then the discharge divided by the area corresponding to normal depth.

4. Choice of protection method

Having calculated the exit velocity and tail water depth an estimate must be made of the scouring that will occur just downstream of the culvert. Methods for doing this can be found in the Culvert Manual and in Hydraulic Handbooks. Should the scour hole be too large for the safety of the culvert then a suitable protection method must be decided upon.

The Froude number based on the culvert outlet flow is an important parameter in selecting appropriate protection. By definition the Froude Number Fr is given by

$$Fr = V/\sqrt{gy} \quad (15)$$

Where V is the mean flow velocity and y , for a rectangular section, is the end depth y_e . For a circular culvert flowing part full an equivalent depth y^* is defined as

$$y^* = (A/2)^{1/2} \quad (16)$$

where A is area of flow at the end and y^* is used in Equation (15).

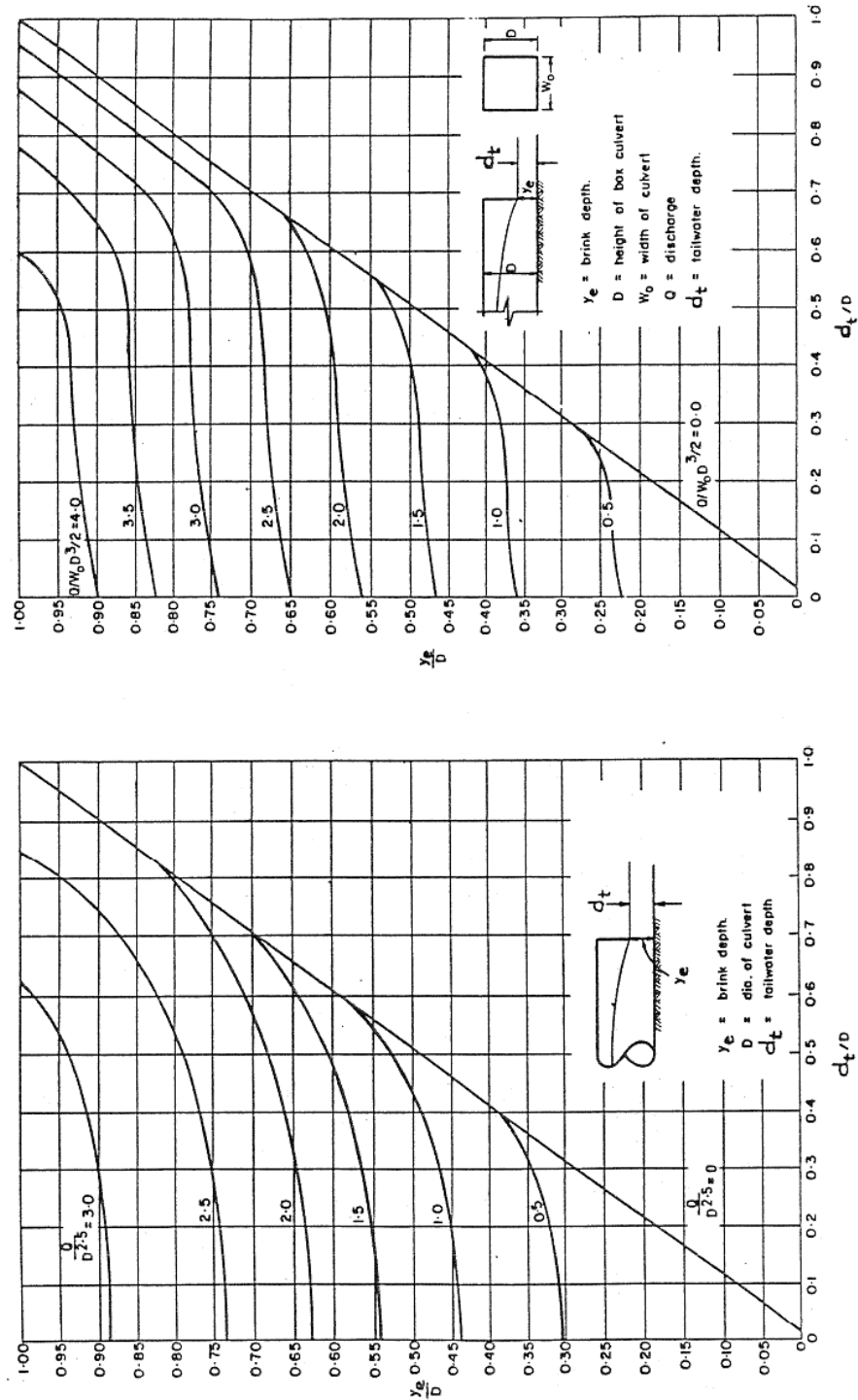


Figure 21. Dimensionless rating curves for culvert outlets on mild slopes

TABLE 3**UNIFORM FLOW IN CIRCULAR SECTIONS FLOWING PARTLY FULL**

y = depth of flow		Q = discharge in cubic metres per second by Manning's formula		
D = diameter of pipe		n = Manning's coefficient (0.012 for concrete pipe)		
A = area of flow		S = slope of the channel bottom and of the water surface		
R = hydraulic radius				
$\frac{y}{D}$	$\frac{A}{D^2}$	$\frac{R}{D}$	$\frac{Qn}{D^{8/3} S^{1/2}}$	$\frac{Qn}{y^{8/3} S^{1/2}}$
0.01	.0013	.0066	.00005	10.31
.05	.0147	.0325	.00150	4.41
.10	.0409	.0635	.00651	3.02
.15	.0739	.0929	.01515	2.38
.20	.1118	.1206	.0273	1.99
.25	.1535	.1466	.0427	1.72
.30	.1982	.1709	.0611	1.51
.35	.2450	.1935	.0820	1.35
.40	.2934	.2142	.1051	1.210
.45	.3428	.2331	.1299	1.092
.50	.3927	.2500	.156	.990
.55	.4426	.2649	.182	.900
.60	.4920	.2776	.209	.818
.65	.5404	.2882	.236	.744
.70	.5872	.2962	.261	.676
.75	.6319	.3017	.284	.613
.80	.6736	.3042	.305	.553
.85	.7115	.3033	.321	.496
.90	.7445	.2980	.333	.440
.95	.7707	.2865	.335	.384
1.00	.7854	.2500	.312	.312

In terms of the Froude numbers as defined above for

$Fr < 1.7$ if a structure is required then a simple apron, rip rap or flow expansion structure will suffice

$1.7 < Fr < 3.0$ A rip rap basin or horizontal roughness element basin is needed.

$3.0 < Fr$ An hydraulic jump basin is needed

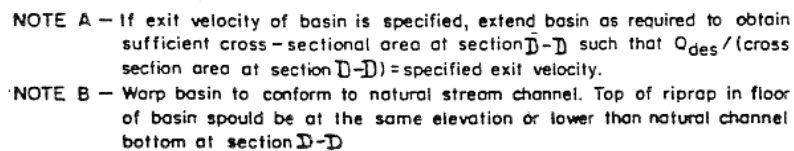
Other considerations in selection include the amount and type of sediment expected to be in transport, the presence or absence of significant tail water depths (tail water must be present for any structure involving an hydraulic jump) and the downstream channel width and alignment.

5. Design of Rip Rap Basin

A rip rap basin is appropriate for $1.0 < Fr < 3.0$. Model studies and field tests have led to a design method in which the depth h_s , length L_s and width W_s of the basin are functions of the culvert outlet Froude Number and the ratio of rip rap size to end depth d_{50}/y_c . Details of the recommended basin shape are shown in Figure 22. The main features are

- (i) The basin is pre-shaped and lined on its base and sides with rip rap of prescribed size.
- (ii) The depth of the basin is set approximately at the depth of scour that would occur in a bed of the rip rap material at the design discharge
- (iii) Energy dissipation is best when h_s/h_{50} lies between 2 and 4.
- (iv) The length of the dissipation pool is the large of $10h_s$ or 3 times the culvert width and the length of the downstream apron is the larger of $5h_s$ or the culvert width.
- (v) The base width increases linearly downstream from W_o , the culvert width to $W_o + (2/3)(\text{the total basin length})$.

The tailwater depth/end depth ratio influences the mode of action of the basin. For $d_t/y_c < 0.75$ the flow from the culvert will plunge into the dissipation pool. This condition should be sought by raising or lowering the culvert invert. If this cannot be achieved and $d_t/y_c > 0.75$ the flow will retain its jet like character through the basin. The jet will diffuse as it goes but some protection downstream of the basin may still be required. Flow velocities in this region then need to be checked and appropriate lining material placed. Velocities can be estimated for different distances from the culvert outlet using Figure 23 and suitable material selected using Figure 24.



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Design procedure

- Determine tailwater depth, end depth y_c , outlet velocity and the value of y ($=y_c$ for a box culvert: $= (A/2)^{1/2}$ for a circular culvert).
- Calculate the Froude number at culvert outlet using Equation (15) and confirm that a rip rap basin is appropriate i.e. $1.0 < Fr < 3.0$.
- Select an available rip rap size d_{50} such that $0.25 < d_{50}/y < 0.45$. This range gives the most satisfactory performance. Large values can be used if necessary.
- Use Figure 25 to determine h_s/y and check that $2 < h_s/d_{50} < 4$. If this is not so then try another value of d_{50}/y .
- Size the basin as shown in Figure 22.
- If $d_t/y_c < 0.75$ and a specified exit velocity is required from the basin, extend its length until the cross sectional area at D-D (Figure 22) equals the design discharge divided by the specified velocity. The depth to be assumed at D-D is the flow depth in the downstream channel.
- If $d_t/y_c > 0.75$ then design a rip rap extension to the basin so that the velocity when the flow reaches the channel is less than the permissible velocity. Use Figures 23 and 24.

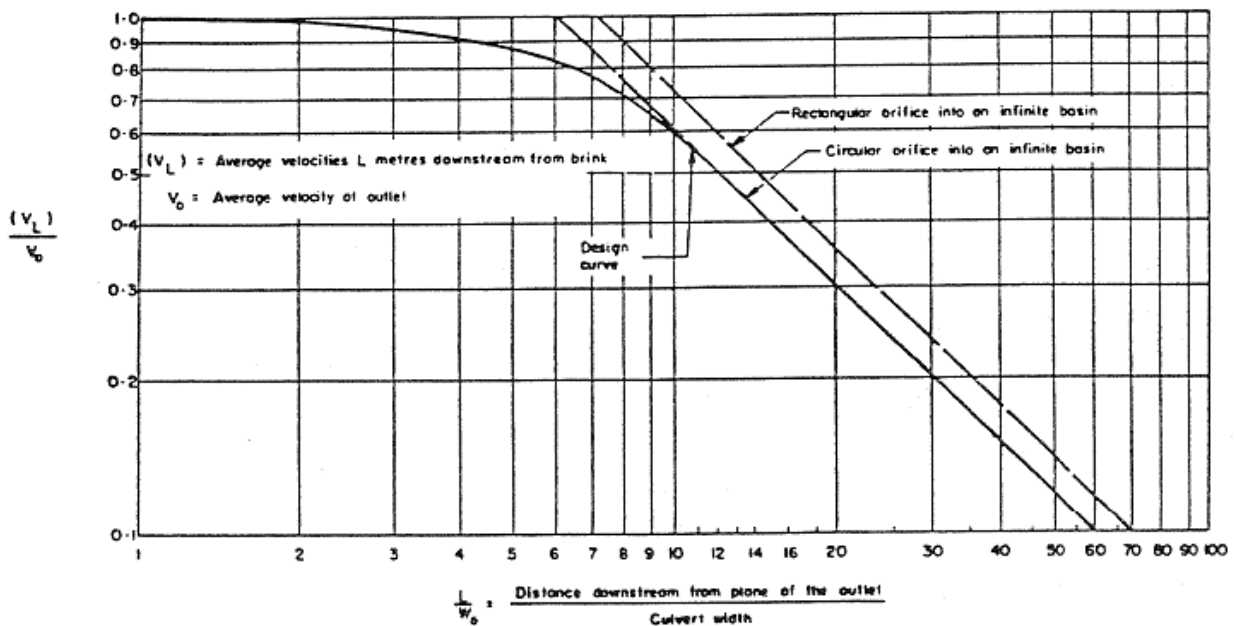
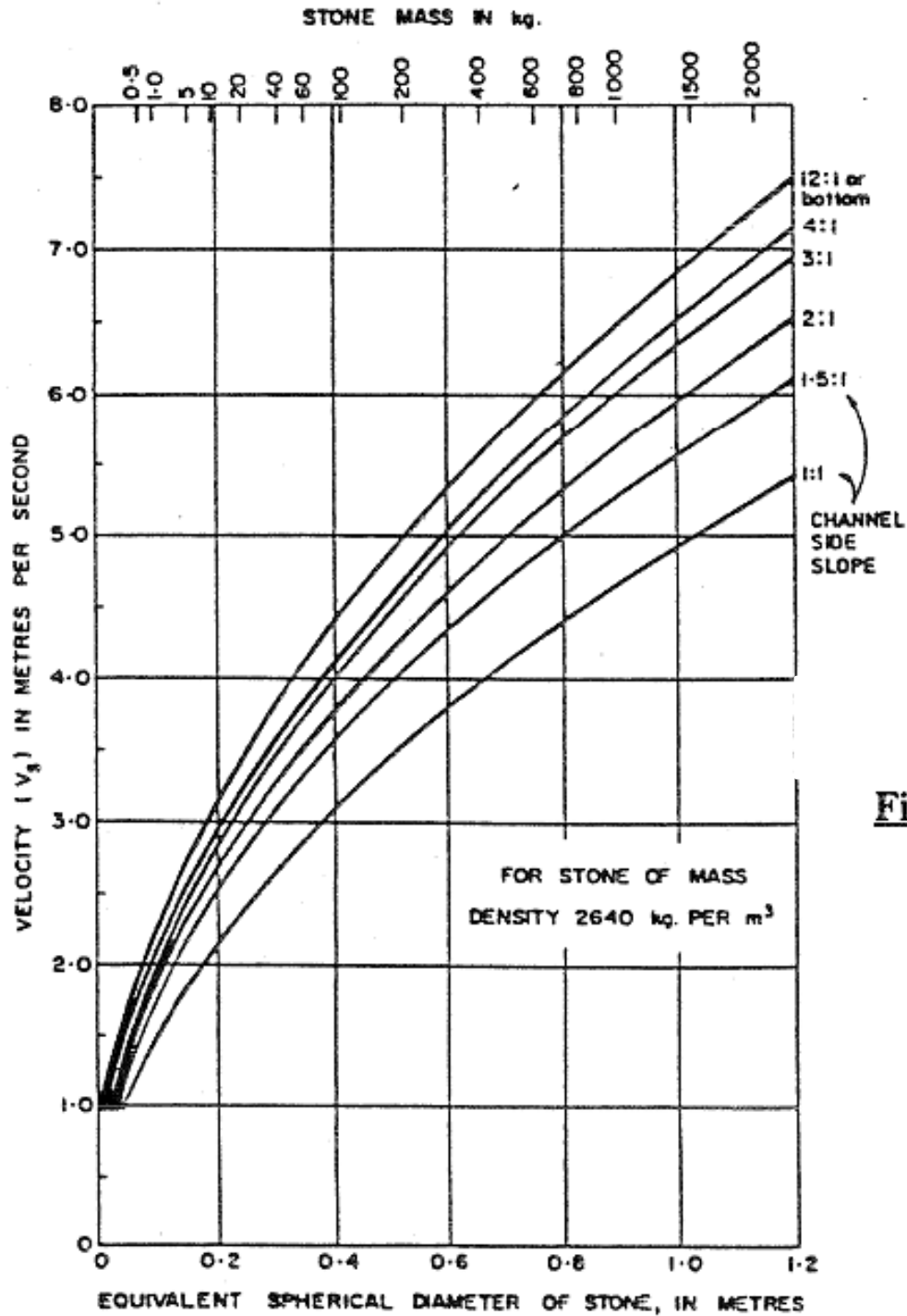


Figure 23. Centre-line velocity as function of distance from culvert outlet ($d_t/y_c > 0.75$)



Fi

Figure 24. Stone sizes required for channel lining

$$V_o = \frac{\text{Design discharge } Q}{\text{Wetted area at brink of culvert}}$$

d_{50} = The median size of rock
Rounded rock or angular rock.

y_e = End depth for box culvert
= $\left(\frac{A}{2}\right)^{1/2}$ For non-rectangular
Section

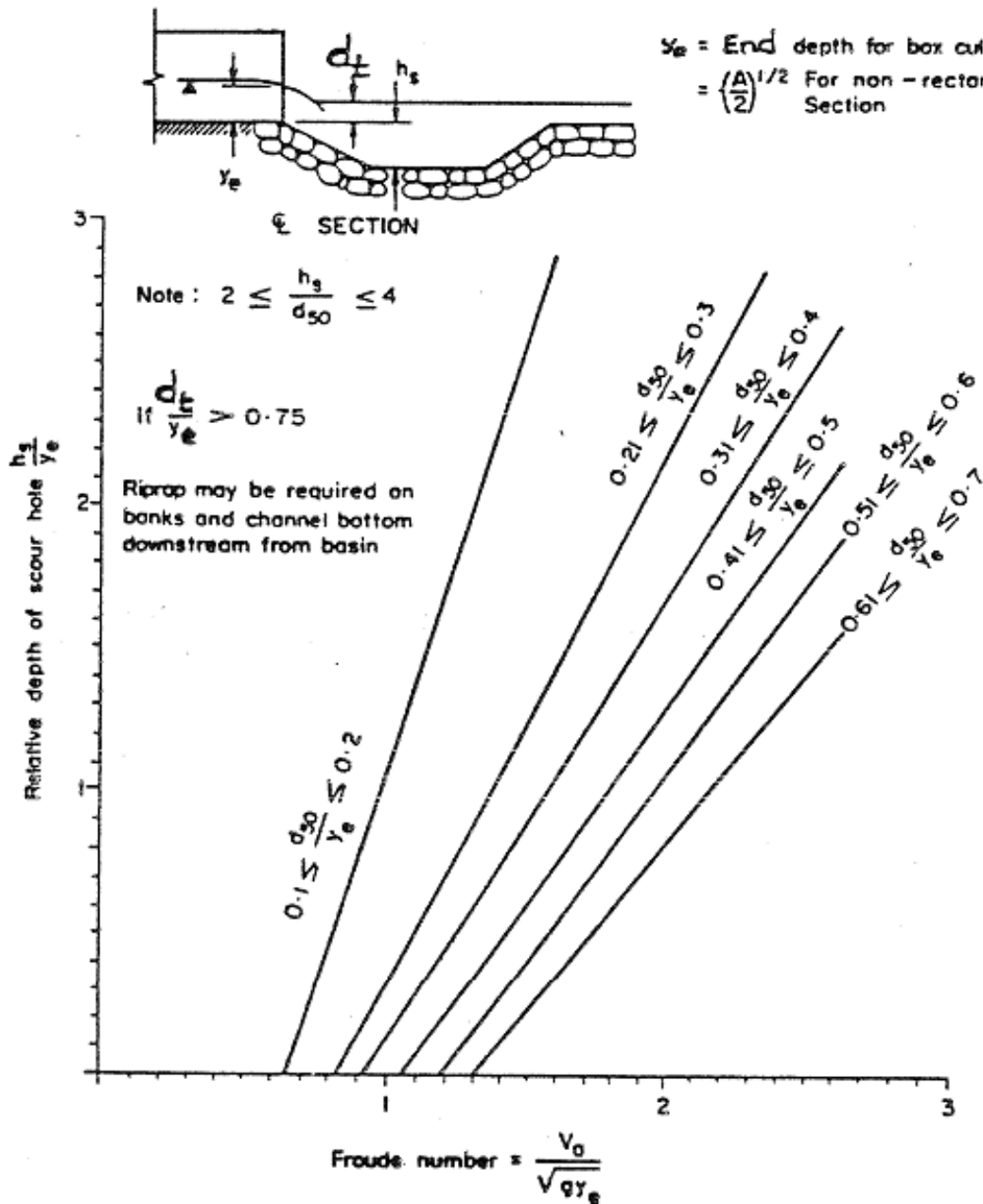


Figure 25. Relative scour hole depth as a function of Froude Number and rip rap size over brink depth

6. Roughness – Element basin

6.1. In a roughness – element basin the flow from the culvert is directed on to a series of blocks or baffles as shown in Figure 26. The blocks offer resistance to the flow and also spread it out to the width of the downstream channel. Design for such a basin is based on application of the momentum principle and model studies to determine drag coefficients for particular configurations of roughness elements.

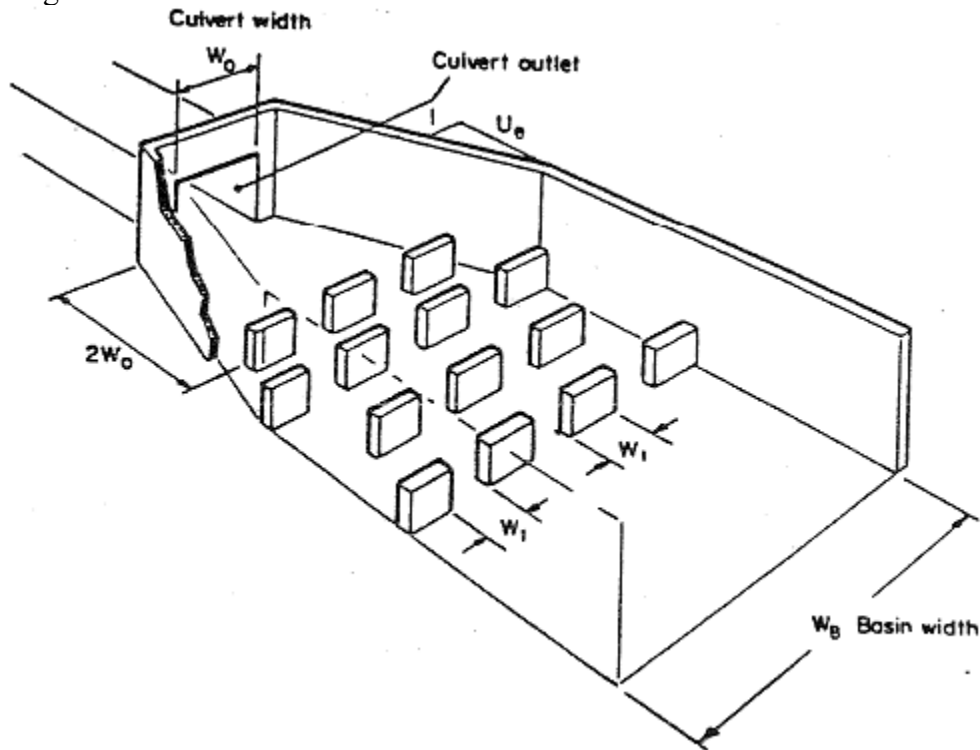


Figure 26. Typical roughness element basin (Source: Culvert Manual)

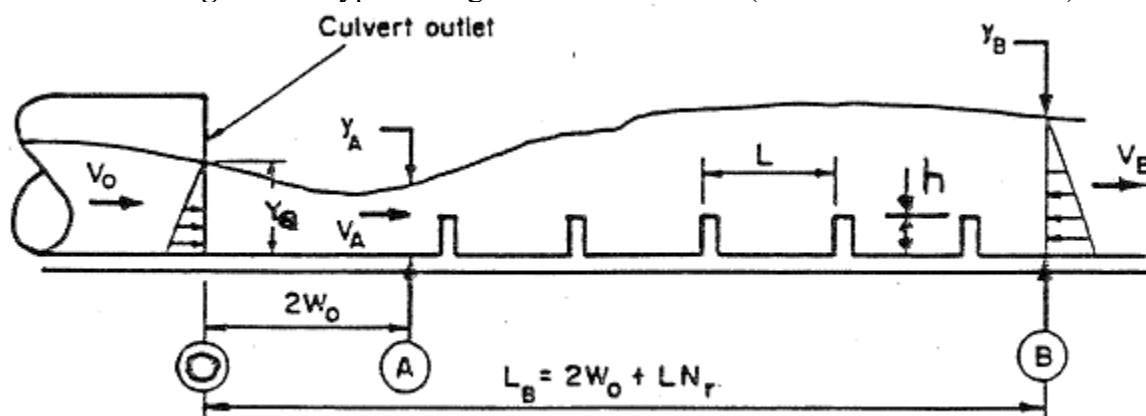


Figure 27. Control volume for momentum equation (Source: Culvert Manual)

7. Analysis

Writing a momentum equation for the control volume OB (Figure 27) we have, for the positive direction assumed from O to B,

$$\alpha \frac{1}{2} \gamma y_0^2 W_0 - \frac{1}{2} \gamma y_B^2 W_B - F = \rho Q (V_B - V_0) \quad (17)$$

Where $\gamma (= \rho g)$ is the specific weight of the fluid

W_0 is the culvert width (or diameter)

y_0 is the end depth y_c

α is a correction factor used in calculating the pressure component at the culvert exit

F is the force exerted on the fluid by the roughness elements.

The correction factor $\alpha = 0.7$ for rectangular culverts and is a function of $Q/(D^2 \sqrt{gD})$ for circular culverts viz:

$$\alpha = 0.56 \text{ if } Q/(D^2 \sqrt{gD}) < 0.67 \quad (18)$$

$$\alpha = 1.0 \text{ if } Q/(D^2 \sqrt{gD}) > 1.0$$

For intermediate values linear interpolation is used.

A drag coefficient C_B is used to relate F to the flow velocity at section A and the total frontal area of the elements

$$F = \frac{1}{2} C_B \rho V_A^2 N W_1 h \quad (19)$$

Where the N elements each have width W_1 and height h .

Values for C_B are given in Figure 28 for different basin geometries and culvert shapes.

The velocity at section A is calculated for a circular culvert from

$$\frac{V_A}{V_0} = 1.65 - 0.45 Q / (D^2 \sqrt{gD}) \quad (20)$$

And for a rectangular culvert from

$$\frac{V_A}{V_0} = 1.65 - 0.35 Fr \quad (21)$$

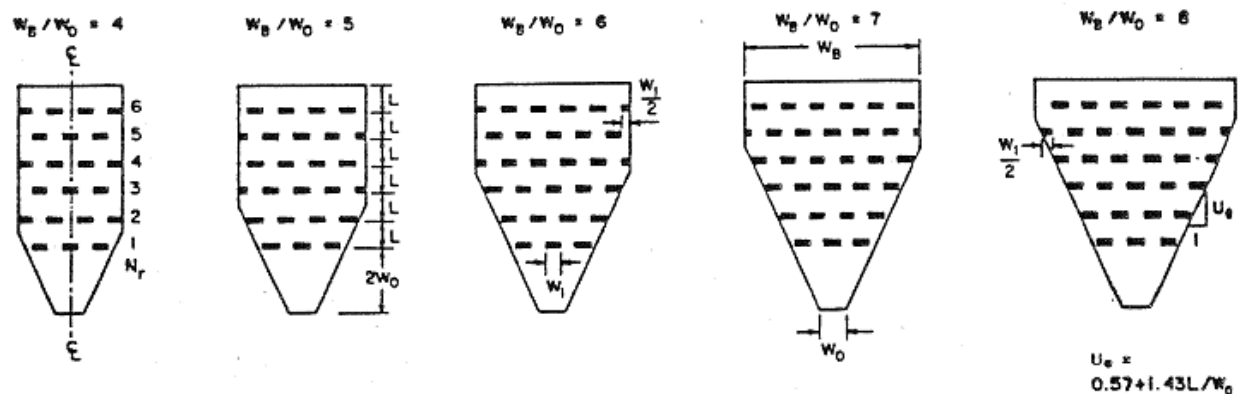
The corresponding depths y_A are given by, for a circular culvert

$$\frac{y_A}{y_c} = 0.21 Fr - 0.04 \quad (22)$$

And for a rectangular culvert

$$\frac{y_A}{y_c} = 0.15 Fr + 0.05 \quad (23)$$

Where the Froude number $Fr = V_0 / \sqrt{g y_c}$



W_B/W_0		2 to 4			5			6			7		8
W_1/W_0		.57			.63			.6			.58		.62
No. Rows (N_r)		4	5	6	4	5	6	4	5	6	5	6	6
No. Elements (N)		14	17	21	15	19	23	17	22	27	24	30	30
h/y_A L/h		C_B FOR ROUGHNESS ELEMENT DISSIPATORS											
Rectangular	0.91 6	0.32	0.28	0.24	0.32	0.28	0.24	0.31	0.27	0.23	0.26	0.22	0.22
	0.71 6	0.44	0.40	0.37	0.42	0.38	0.35	0.40	0.36	0.33	0.34	0.31	0.29
	0.48 12	0.60	0.55	0.51	0.56	0.51	0.47	0.53	0.48	0.43	0.46	0.39	0.35
	0.37 12	0.68	0.66	0.65	0.65	0.62	0.60	0.62	0.58	0.55	0.54	0.50	0.45
Circular	0.91 6	0.21	0.20	0.48	0.21	0.19	0.17	0.21	0.19	0.17	0.18	0.16	
	0.71 6	0.29	0.27	0.40	0.27	0.25	0.23	0.25	0.23	0.22	0.22	0.20	
	0.31 6	0.38	0.36	0.34	0.36	0.34	0.32	0.34	0.32	0.30	0.30	0.28	
	0.48 12	0.45	0.42	0.25	0.40	0.38	0.36	0.36	0.34	0.32	0.30	0.28	
	0.37 12	0.52	0.50	0.18	0.48	0.46	0.44	0.44	0.42	0.40	0.38	0.36	

Figure 28. Design values for roughness element dissipators (Source: Culvert Manual)

8. Design procedure

- Determine outlet conditions for the culvert, y_c , V_0 and calculate $Fr = V_0/\sqrt{gy_c}$. If $Fr < 3.0$ then a roughness element basin can be used.
- Select a basin from Figure 28 that fits the site geometry viz. a suitable value for W_B and thus W_B/W_0 and hence W_1 .
- From Figure 28 select N_r , and h/y_A to obtain a value of C_B .
- Determine V_A and y_A from Equations (20) to (23) and thus h , L and $L_B = 2W_0 + LN_r$.
- Compute the force F using the momentum equation. Equation (17). Use tailwater depth for y_B .
- Calculate F from Equation (19) using the values determined in 2, 3, 4 above. If this value exceeds that found from the momentum equation the design is satisfactory. If not return to 2 and select a different basic geometry.
- Finalize layout of basin by noting
 - Placement of elements is to be symmetrical about the centerline
 - Lateral spacing of elements should approximate the element width W_1
 - Check W_1/h lies between 2 and 8

- Stagger the rows and ensure at least half the rows have elements near the basin walls.
- (viii) Determine the need for any riprap protection downstream of the basin by reference to Figure 24.

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